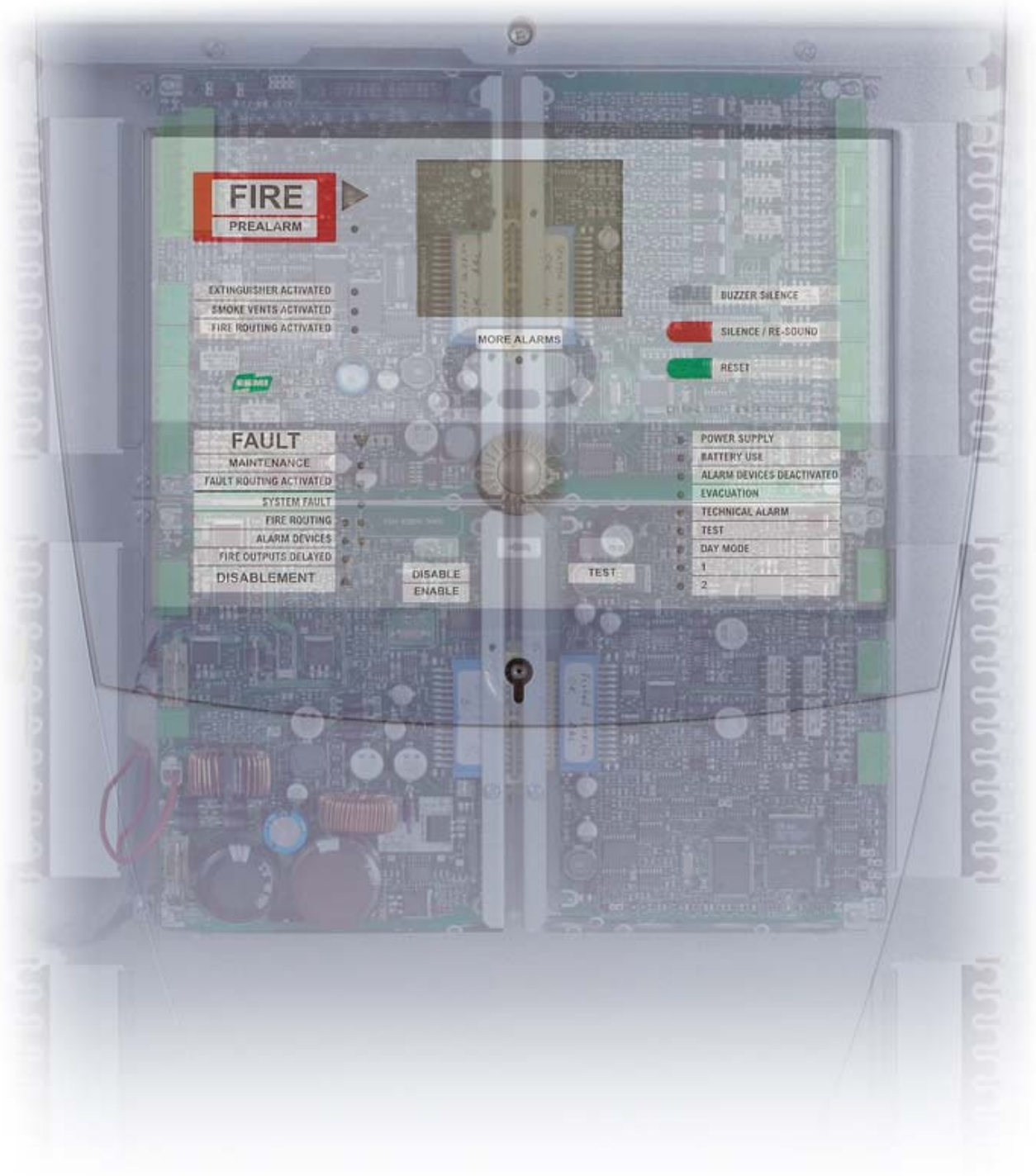
FX FIRE DETECTION SYSTEM

Installation and commissioning manual



Read this manual carefully before installation and commissioning!
Installation and commissioning must be performed according to this manual.
This Installation and Commissioning Manual is to be kept together with the control panel.

6657 1478GB4 19-2006

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About this document

For non-experienced installers this document provides sufficient information to install the FX panel and to commission the whole system successfully.

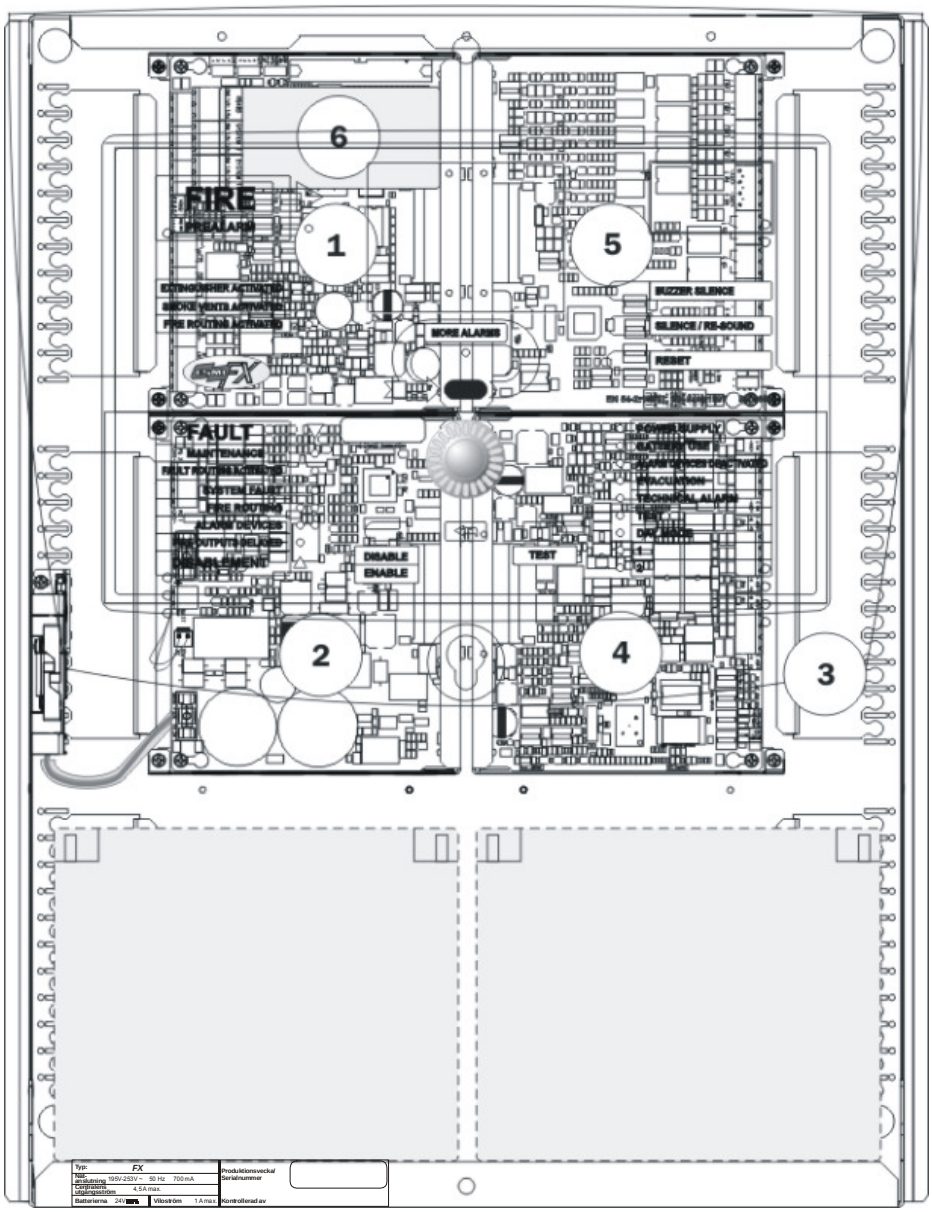
Experienced installers may select to connect all loops and IO connections at once without testing the connections in between.

However, it is strongly recommended always to do the preliminary checks and an initial test run before any cables (except the necessary main supply) are connected to the panel

This document uses the following acronyms for the various units in the panel:

1 MC – Master Controller
2 PS – Power Supply
3 UI – User Interface
4 LC – Loop Controller
4 CLC – Conventional Line Controller
4 ALC - Loop Controller
5 IOC - IO Controller
6 SAA, SAB – Serial Adapter

Typical placement of the units in a panel



1. Installation

The mounting surface must be flat and it must bear the weight of the control panel and the chart file cabinet.

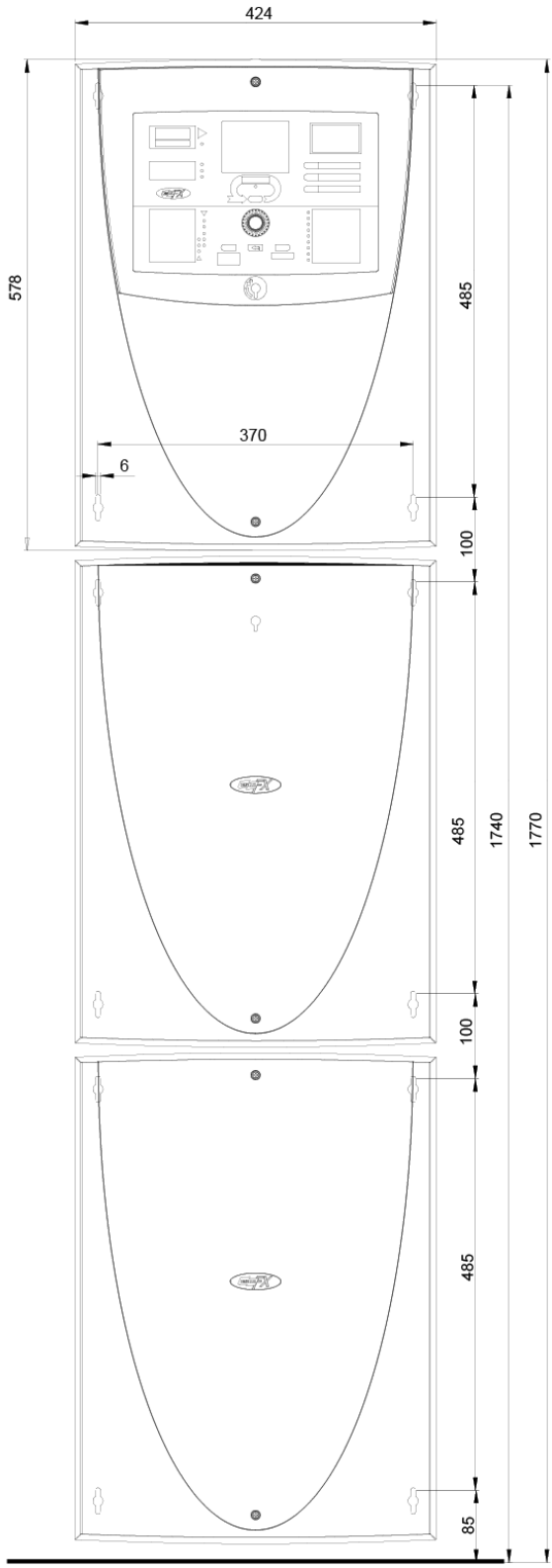
The weight of the control panel excl. batteries is 11 kg. The weight of the chart file cabinet is 5 kg.

The mounting is to be made straight to the wall surface, without any distance bushings or similar, to ensure ingress protection class of IP20.

FX control panel

Chart cabinet

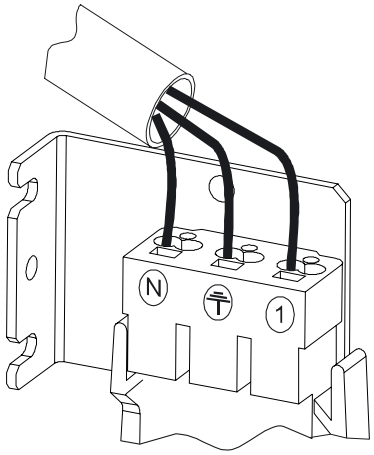
Battery cabinet



2. Connecting main power supply

Connect the mains supply (230 VAC)

- There must be a separate group fuse (10A) for the control panel power supply.
- Cable 3 x 1.5 mm².



3. Commissioning

3.1 Necessary devices and documents

Devices

- A universal measuring instrument (voltage, current, resistance, diode).
- A PC and the configuration tool, if the configuration is done when commissioning.

Documents

- This Installation and Commissioning manual.
- Operation manual.
- Planning and installation documents for the project.
- Client/project configuration data if the configuration is done when commissioning.

WARNING!
An isolation resistance meter must not be used for measuring resistance.

Note!
The system does not require configuration in order to function.

On the other hand, client specific features may require configuration.

3.2 Order of commissioning

1. Check that the installation has been done correctly according to the plans.
2. Make preliminary checks on the control panel.
3. Test run the control panel.
4. Connect detector loops.
5. Connect alarm device lines.
6. Make site specific settings.
7. Configure the system if required for the plan or site specific features.
8. Connect the outputs.
9. Connect the inputs.
10. Connect the serial communication ports.
11. Connect the router to the control panel.

Note!
For LC only (System Sensor devices)
If a loop has the same addresses for both detectors and I/O-modules, and the system is not configured, the detectors will be assigned addresses from the low range and the I/O-modules from the high range (e.g. 001 and 101).

Usually it is easier to connect the loop, if the control panel is not configured.

Note!
On ALCA / ALCB - loop (Intellia devices) all detectors and modules must have individual addresses from 1 to 126.

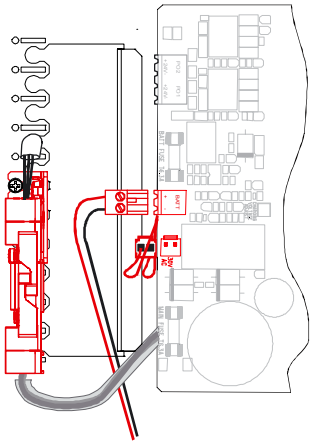
4. Preliminary checks

4.1 General

The aim of the preliminary checks is to assure that the settings are correct and that the control panel has not suffered any defects during transportation or installation. This is obviously most easy to do when no external cables are connected, except for the necessary power supply connection.

4.2 Preliminary checks

1. Check that all power is disconnected from the control panel.
   - The cable from the transformer to the power supply component board is disconnected from the terminal 30VAC on the component board.
   - The battery cable is disconnected from the battery terminal BATT on the component board.

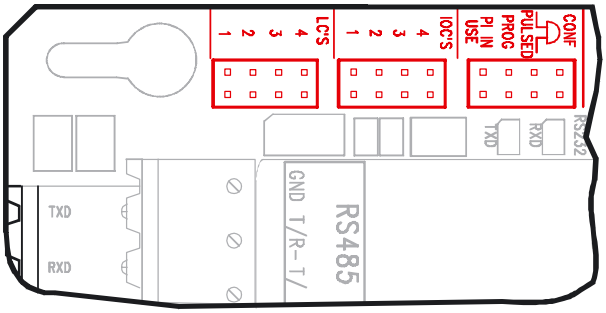


2. Check that the mains cable is connected to the mains terminal.

3. Check that the separate group fuse reserved for the control panel is in its place.

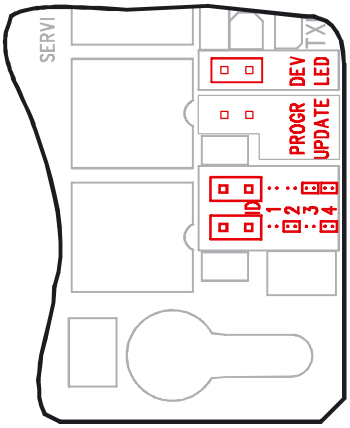
MC jumper settings

1. Check the following settings:
   - The “CONF” jumper is not in place.
   - The “PULSED” jumper is not in place (for continuous alarm device signal) or in place (for pulsing alarm device signal).
   - The “PROG” jumper is not in place.
   - The “PI IN USE” jumper is not in place.
   - The “IOC’S” jumper is correctly set for the number of IOC’s in the panel.
   - The “LC’S” jumper is correctly set for the number of LC’s, ALC’s and CLC’s in the panel.
2. Check also that there is a 4.7 kΩ end-of-line resistor in the terminals for the alarm device line.



LC jumper settings

1. Check the following settings:
   - “Prog Update” jumper is not in place.
   - “Dev LED” jumper is in place if you want the detector leds to flash when communicating with the panel or not in place otherwise.
   - The LC address setting is different for each LC, ALC and CLC and ranging from 1 to the number of LC’s, ALC’s or CLC’s.

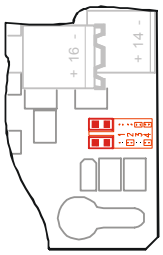


2. Check also that there is a wire between A+ and B+ as well as between A- and B- in the terminals for both loops.

CLC jumper settings

1. Check the following settings:

- The LC address setting is different for each LC, ALC and CLC and ranging from 1 to the number of LC's, ALC's or CLC's.
- Check also that all conventional line terminals have their EOL resistors connected to the terminals.

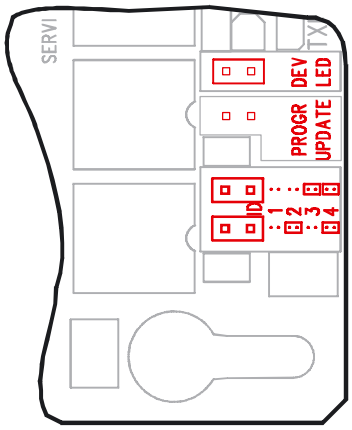


ALC jumper settings

1. Check the following settings:

- "Prog Update" jumper is not in place.
- "Dev LED" jumper is in place if you want the detector leds to flash when communicating with the panel or not in place otherwise.
- The LC address setting is different for each LC, ALC and CLC and ranging from 1 to the number of LC's, ALC's or CLC's.

2. Check also that there is a wire between A+ and B+ as well as between A- and B- in the terminals for both loops.

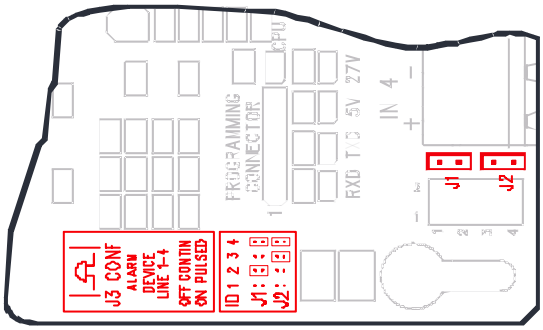


IOC jumper settings

1. Check the following settings:

- The IOC address setting is different for each IOC and ranging from 1 to the number of IOC's.
- The alarm device line configuration jumpers are set for desired operation (pulsed or continuous signal).

2. Check also that there are 4,7kΩ end-of-line resistors in the terminals for each alarm device line.

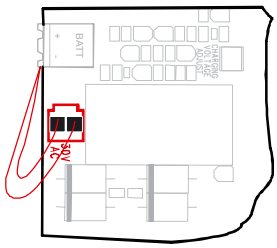


5. Control panel test run

5.1 Mains connection

Note! The panel is in access level 2 when the cover is removed.

1. Switch the panel on by connecting the cable between the transformer and the terminal 30VAC on the power supply component board.



- The backlight of the display flashes for about 20 s.
- The display will show the text:
- The text "Panel starting up" will disappear from the display when all addresses are scanned for presence.

13:36	**11.11.2004**
FX fire panel	
Activate menu by pressing the wheel	
Panel starting up	

- After approximately one minute from start the panel will indicate a fault alarm due to the missing battery:

Buzzer beeps continuously
FAULT-led blinks

- LCD display shows:

FAULT	1/1
PS battery fault	
Panel starting up	

2. Press the “BUZZER SILENCE”- button

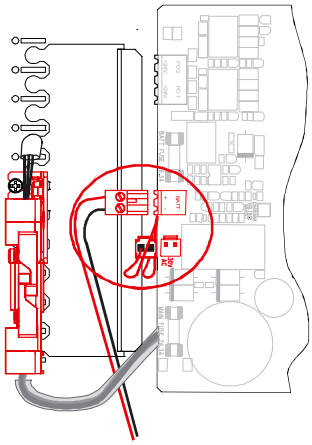
The audible signal is silenced
The “FAULT”- led becomes fixed.
The LCD remains the same

The panel will also indicate a fault alarm for missing configuration data if the panel is not configured.

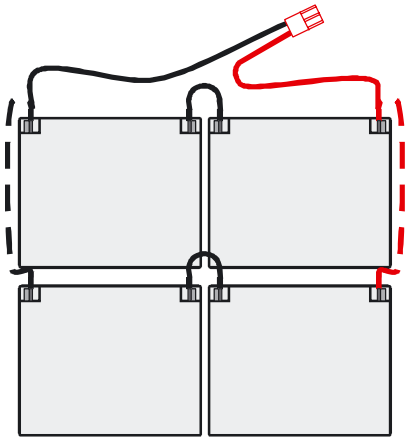
This fault alarm can be reset and will not appear again.

5.2 Battery connection

1. Install the batteries in the cabinet. Check that the battery cable is disconnected from the battery terminal BATT (*Fig 2*).



2. Check battery polarity from the markings on the battery
3. Connect the battery cables to the poles as follows:
   - The batteries are 12 V batteries. If two batteries are used, they are connected in series.
   - If four batteries are used, two series connected pairs are connected in parallel (Voltage is 24 VDC).



4. Connect the battery cable to the battery terminal BATT on the PS component board.

WARNING!

Connecting the batteries in the wrong way may cause a short circuit in the batteries, which may lead to an explosion, a strong electric arc or fire in the battery cables.

5. Reset the battery fault indication in the control panel by pressing the “RESET”- button until the pulsed audible signal stops.

The “FAULT”- led goes out The display shows the text:
FX fire detection system
Activate menu by pressing the wheel

6. Cable handling and preliminary measurements

WARNING!
Check by measuring that there is no power in the cables.

6.1 Cable handling and measurements

- At every stage of commissioning, power must be disconnected from the control panel before the cables are installed.

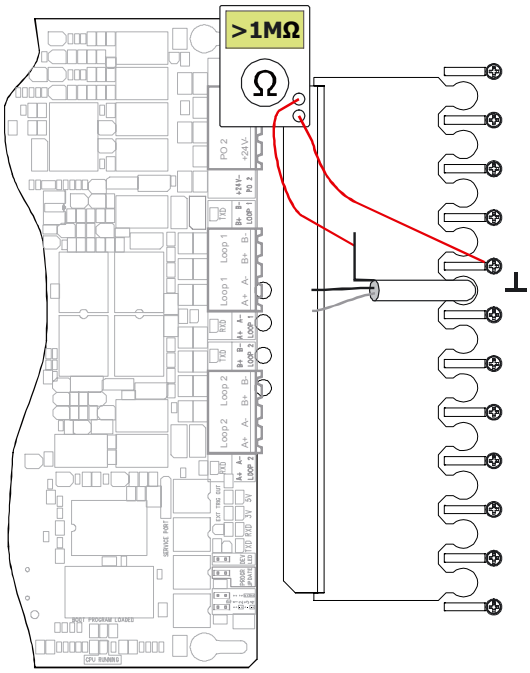
1. Peel off the cable plastic and protective shields. Be careful not to let peeling remains drop onto the component board or between the back wall of the cabinet and the component board.
2. Make sure that the cables have been marked and that the markings can be seen after the peeling.
3. If the cables used are shielded, measure, before connecting the shield that the resistance between the shield and the cabinet potential equalization bar exceeds 1 MΩ . If the resistance is smaller, the cable shield is in contact with the construction of the building:

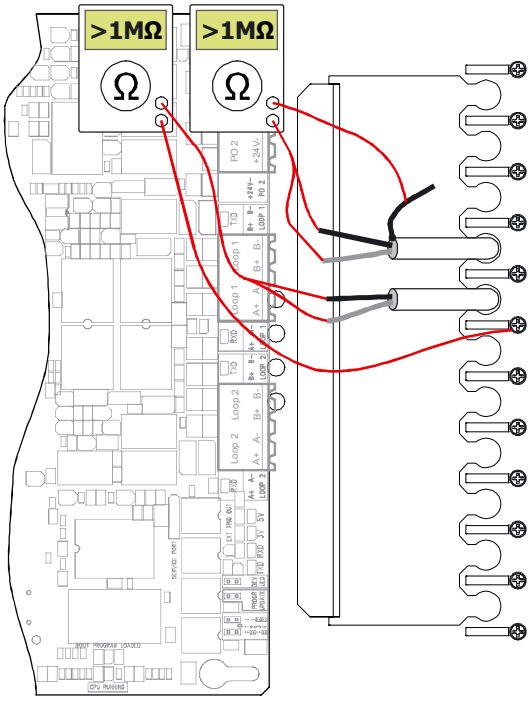
- The outer sleeve of the cable has been damaged.
- The protective shield is in connection to the construction of the building in a detector socket.

  Repair the fault. Then connect the cable shield to the frame of the control panel.

4. Measure for each cable: the resistance between the cable shield and the wires. The resistance must exceed 1 MΩ . If the value is smaller, the earth leak must be located and repaired.
5. If the cables used are not shielded, measure the resistance between the wires and the control panel framework. It must also exceed 1 MΩ.

First disconnect the battery then the mains.





6.2 Cable table

Cable connection	Conductors x area	Max. length	Comments
Addressable detection circuit cables, LC - loop	2 x 0,5 mm^2 + shield 2 x 1,0 mm^2 + shield	540 m (50 Ω) 1000 m (40 Ω)	The cable resistance of the loop is max. 40Ω and the capacitance max. 180nF between conductor and shield, 360nF between conductors. Max. voltage drop is 8V
Addressable detection circuit cables, ALC- loop	2 x 0,5 mm^2 + shield 2 x 1,0 mm^2 + shield	810 m (60 Ω) 1500 m (60 Ω)	The cable resistance of the loop is max. 60Ω and the capacitance max. 180nF between conductor and shield, 360nF between conductors. Max. voltage drop is 9V
Conventional detection circuit cables, CLC- line	2 x 0,5 mm^2 + shield 2 x 1,0 mm^2 + shield	1200 m (100 Ω) 2400 m (100 Ω)	The cable resistance of the loop is max. 50Ω, if an Exi barrier is connected to the loop, otherwise max 100Ω. The max. allowed capacitance of the cable is 0.5µF.
Sub-detection circuits of conventional zone modules, LC- loop	Sub-detection circuits of conventional zone modules	1200 m (100 Ω)	Conventional zone module EM210E-CZ/M512ME and conventional detectors or conventional manual call points
Sub-detection circuits of conventional zone modules, ALC- loop	Sub-detection circuits of conventional zone modules	600m (50 Ω)	Conventional zone module EMI-310-CZ / EMI410-CZ and conventional detectors or conventional manual call points
Power supply to conventional zone modules	2 x 0,5 mm^2 + shield 2 x 1,0 mm^2 + shield	625 m (50 Ω) 1200 m (50 Ω)	Cable resistance max. 50Ω
Conventional Eex-area sub-loop unit	2 x 0,5 mm^2 + shield 2 x 1,0 mm^2 + shield	625 m (50 Ω) 1200 m (50 Ω)	From panel through the address module to the Eex barrier resistance max. 50 Ω total. If power supply is brought to several units through the same cable, the real length of this part must be multiplied with the number of units when comparing with the maximum length.
	2 x 0,5 mm^2 + shield	150 m	Loop resistance from the Eex-barrier to most distant detector max.12Ω. Loop capacitance max.390nF.
Printer connection - Serial data	5 x 0,5 mm^2 + shield	15 m	RS232
Serial connections - INFO - MESA/FX	2 x 0,5 mm^2 + shield or 2 x 0,5 mm^2	1000 m	RS485
FX clean contact input lines	2 x 0,5 mm^2	2000 m	
FX clean contact output lines	2 x 0,5 mm^2 or 2 x 1,0 mm^2	To be calculated separately	The equipment receiving the contact signal may have restrictions on cable properties. Load controlled by the relay output may restrict allowed resistance and length per cross section. Relay contact rating is 30Vdc, 1A
FX alarm device lines – fire bell, siren line - fault buzzer line	2 x 0,5 mm^2 or 2 x 1,0 mm^2 or 2 x 2,5 mm^2	To be calculated separately	Max. allowed voltage drop defines cable to be used.
Addressable monitor modules, LC- loop - monitor lines	2 x 0,5 mm^2 + shield	625 m (50 Ω)	Monitor modules M500ME, M503ME, M501ME EM210E, EM220E, EM221E
Addressable control modules, LC- loop - power supply - alarm line	2 x 0,5 mm^2 or 2 x 1,5 mm^2 or 2 x 2,5 mm^2	To be calculated separately	Control modules M500CHE, EM201E and EM221E Number and distances of the relay control modules define the conductor area and length of the power supply cable.
Addressable monitor modules, ALC -loop - monitor lines	2 x 0,5 mm^2 + shield	625 m (50 Ω)	Monitor modules EMI- 310, EMI-310+, EMI-311, EMI-311/240, EMI-333, -EMI-410, EMI-410+, EMI-411 Mini monitor modules 55000-833 APO, 55000-832 APO
Addressable control modules, ALC- loop - power supply - alarm line	2 x 0,5 mm^2 or 2 x 1,5 mm^2 or 2 x 2,5 mm^2	To be calculated separately	Controll modules EMI-301, EMI-311, EMI-311/240, EMI-301S, EMI-401, EMI-401S,EMI-411
Mains supply cable	3 x 1,5 mm^2		Mains connection: - 230 ±10% V AC, 50-60 Hz, - maximum power 160 W - own circuit fuse 10 A

NOTE!
The max. current consumption of the **FX** - control panel, all loops and addresses and all control panel outputs is 1 A in standby condition and 4.5 A in alarm condition. The standby time required for the system may limit the max. load of the outputs.

NOTE!
The max. current consumption of the **FXM** - control panel (with PSA- power supply board), all loops and addresses and all control panel outputs is 0.5 A in standby condition and 2.2 A in alarm condition. The standby time required for the system may limit the max. load of the outputs.

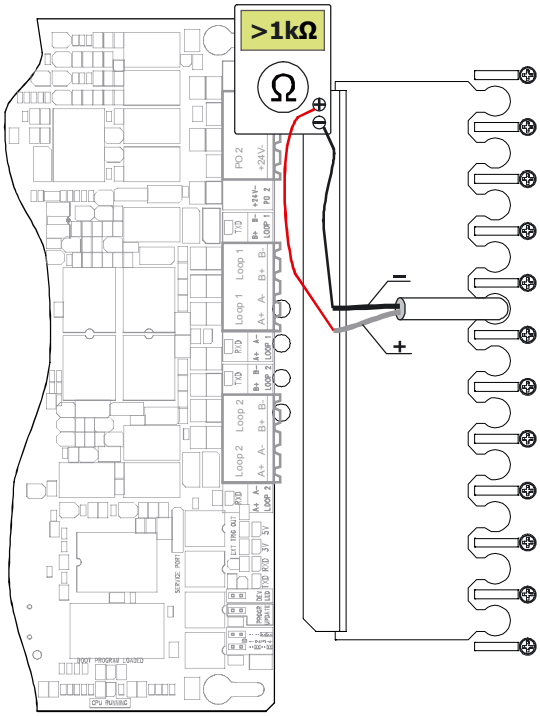
7. Addressable detector loops

7.1 Measuring the Cables

- Disconnect all power from the control panel

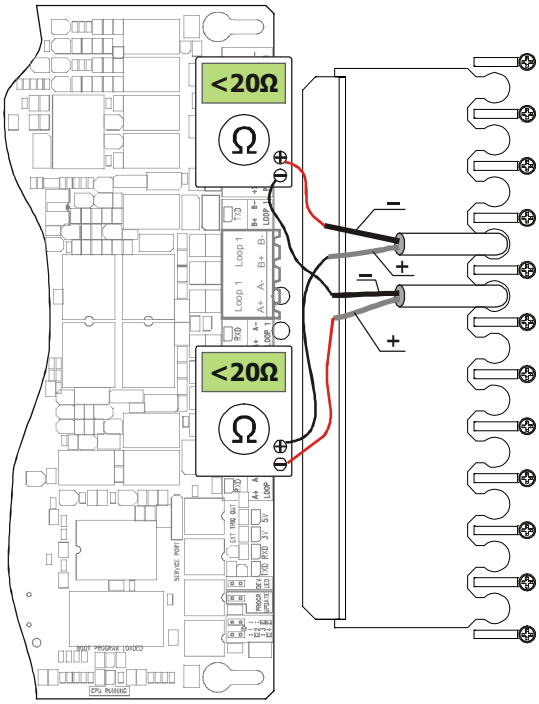
First disconnect the battery, then the mains.

1. Measure the resistance between the loop wires + and - so that the resistance meter positive (feeding) probe is in the + -conductor. The resistance must exceed 1kΩ. If the resistance is smaller, locate the cause of the fault and repair it. (It may be a detector or address unit connected the wrong way or a short circuit isolator).



2. Measure the loop cable + wire resistance from the output and return ends. Also measure the – wire. The loop resistance of the + and – wires added together can be max. <u>40Ω on LC - loop</u> and <u>60Ω on ALC - loop</u>.

The difference between the + and – wire resistances should not exceed 5Ω The possible causes of a fault must be located and repaired.



Note!

If there are short circuit isolators in the loop, resistance measuring of the loop cable will give false results. The short circuit isolators must be by-passed by connecting links over the isolators.

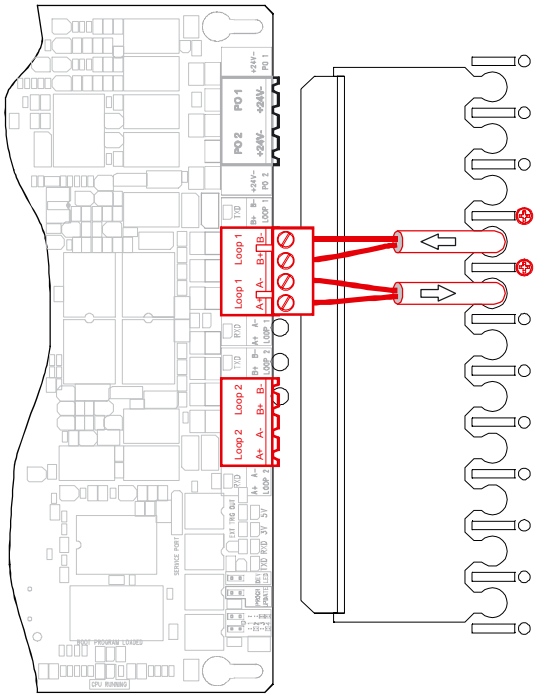
3. Remove short circuit isolators' by-passes.

7.2 Connecting the loop to the control panel

Each FX-LC and –ALCB board has connectors for two loops
ALCA -board has only one loop.

1. Check that all power has been disconnected from the control panel.
2. If shielded cable is used, connect the shield to the nearest earth screw in the back plate.
3. Connect the loop wires to the LOOP terminals of the LC/ALC board.

- Outgoing end to terminals A+ and A-.
- Return end to terminals B+ and B- .



7.3 Functional check of the loop

1. Start the control panel by switching the power on

First connect the transformer cable, then the battery cable

2. Wait for about 3 minutes, until the panel has started up and scanned all devices connected to the loops.
3. Verify in the panel display that all addresses are indeed found by selecting "Loops/Address points" from the menu, and stepping through all addresses in all loops. (This is not necessary if the panel is configured)

The text 'Panel starting up' disappears from the bottom of the display.

The detector loop is in order if the control panel does not give fire, fault or service indications. Otherwise the fault must be located by following the instructions below.

4. If the panel indicates fire alarm, press the "BUZZER SILENCE" button, scroll the alarm indications with the "MORE ALARMS"- button and note them.
5. Eliminate the causes of fire alarm and reset the indications by pressing the "RESET"- button until the pulsed signal stops.
6. Repeat the above mentioned procedures until there are no more fire alarms.

Possible causes for fire alarm:

- A manual call point is pressed down; the glass has been broken or taken off.
- A manual call point or another device connected to the input of an addressable monitoring unit is in active state.
- Smoke, water vapour or thick dust in a smoke detector.
- Heat close to a heat detector.

7. If the panel indicates fault or maintenance warning press the "BUZZER SILENCE" button, scroll the indications with the "MORE ALARMS"- button and note them.
8. Eliminate the causes of fault and maintenance warnings and reset the indications by pressing the "RESET"- button until the pulsed signal stops.
9. Repeat the above mentioned procedures until there are no

Possible causes for fault or maintenance warnings:

- A break or a short circuit in the loop. See below for how to find the fault.
- An earth leak in the loop. See below for how to find the fault.
- An address fault in the loop. See table

more fault and maintenance warnings.

below for possible address fault indications.

A short circuit indication is caused by a loop component connected the wrong way or a short circuit in the wires. If short circuit isolators are used, the shorted part of the loop between the closest isolators will be non-operative. By scrolling trough the addresses list for the shorted loop and comparing with the installation plan, the shorted area can be located. (If the panel is configured, it will indicate fault alarm for all addresses between the operated short circuit isolators).

A break in the loop is easily found by disconnecting the return end of the loop and comparing the addresses that the panel can communicate with, with the installation plan.

An earth fault can most easy be found by splitting the loop in two parts, disconnecting the return end of the loop from the panel and restart. If the panel still indicates earth fault, the location is in the still connected part of the loop, otherwise in the non-connected part. By splitting the faulty part of the loop in two and restarting again, you will soon find the reason for the earth fault.

Fault and maintenance warnings of the loop devices are identified by codes in the display. The codes and a short explanation of each are listed below.

Run time faults:	
FAULT 51:	Too low analogue value (< 400 us) received from analogue sensor Fault in the internal operation of an OMNI / 2251TEM / 7251LASER sensor
FAULT 52:	Invalid reply from the address.
FAULT 53:	Two or more devices have the same address ("double address”).
FAULT 54:	LC-loop: break in the input circuit of a monitor module. Must have 47kΩ EOL- resistor
FAULT 54:	ALC-loop: break or short circuit in the input circuit of a monitor module. Must have 20kΩ EOL-resistor AND 1kΩ serial resistor on the monitored contact
FAULT 55:	Break in the output circuit of a control module.
FAULT 56:	Short circuit in the output circuit of a control module.
FAULT 57:	Input module has been configured as "fault input". When module alarms FX show it as fault warning.
FAULT 58:	Input module has been configured as "zone disablement device" and disablement time exceeds (default 12h).
MAINTENANCE 59:	Input module is configured as "maintenance input". When module alarms FX show it as maintenance warning.
MAINTENANCE 60:	A dirty detector. If the analogue value of the detector has exceeded the maintenance limit for more than 24 hours, this warning is indicated. If detector exceeds value (during 24h) after user reset this fault warning, fault warning is re-generated immediately. Also at case that value of some other detector(s) stay over the maintenance limit (but 24h timeout has not expired) at the moment when user select from the menu “report dirty detector” and press “enable” at access level 3, this fault warning is created. OMNI / 2251TEM / 7251LASER drift compensation alert OMNI / 2251TEM >200 <560; 7251LASER >200 < 650
FAULT 62:	LC- loop: Break or short circuit at the conventional sub loop of conventional zone module. Must have 3,9kΩ EOL- resistor
FAULT 62:	ALC- loop: Break or short circuit at the conventional sub loop of conventional zone module. Must have 6,1kΩ EOL- resistor
FAULT 63:	Invalid response from detector (> 4000 µs for normal sensors, >860 and < 1600 for the OMNI sensor
FAULT 64:	Device type (or functional type) changed when loop is running.
FAULT 65:	LC- loop: sensor with separate remote LED control feature is changed to same type device without this feature. More often this event is indicated with fault 64
FAULT 66:	Input module is configured as "fault in extinguisher ". When module alarms FX show it as fault warning
FAULT 70:	Undefined type of device
FAULT 71:	Memory write of device failed
FAULT 72:	Memory read of device failed
FAULT 73:	Communication fault between ALC and device

FAULT 90:	Unacceptable device
FAULT 92:	Break in input line of the CLC unit
FAULT 93:	Short circuit in input line of the CLC unit
FAULT 94:	Voltage problem in input line of the CLC unit

The following fault and maintenance warnings may be indicated after panel/detection circuit start up (panel re-started, loop controller (LC) re-started or detection circuit has been disable/enabled at access level 3).

When address points are configured with PC:	
MAINTENANCE 00:	The 'any type' of device has been configured for this address but no device is installed. **This is indicated only once and will disappear after resetting the warning.**
MAINTENANCE 01	No type has been configured for this address, but the panel has found an address installed at this address. **This is indicated only once and will disappear after resetting the warning.**
MAINTENANCE 02	Definite type of device has been configured but no device is installed. This warning can be removed only by installing device to this address or changing the configuration.
MAINTENANCE 03	The configured type and installed type is not the same. This warning can be removed only by changing device to this address or changing the configuration.
MAINTENANCE 11	LC-loop: siren control (separate remote LED control) has been configured for the address but the feature is not supported by the device.
FAULT 12	Technical alarm input configured for LC address but SW version of LC is not compatible (>1.3)
FAULT 13	Two or more devices have the same address
FAULT 14	Bad response from the device
FAULT 15	Frame ID is not compatible with memory ID

When there is no configuration for address points:	
MAINTENANCE 04:	LC- loop: a module and a detector have the same address switch setting. This is not allowed without configuration. Module data will replace detector data in memory. Addresses have to be fixed and either the panel has to be restarted or the loop has to be disabled and re-enabled in service state.
MAINTENANCE 05:	LC- loop: during start-up, the panel has found addressable devices with address setting 00. The LED's of the detector or module blink automatically for easier identification. Fire alarm or other fault warnings will not be indicated for units that have the 00 address.
FAULT 06:	Undefined type of device. **This is indicated only once. After resetting the warning the device in this address is removed from data memory.**
FAULT 07:	Device removed from this address during loop disablement. The fault is detected when the loop is re-enabled. Disablement of an address with this fault is not possible because the data of the unit will be removed from memory when the fault is reset. If disabled before the loop enablement, the disablement will also be removed automatically when the fault is reset. **This is indicated only once and will disappear after resetting the warning.**
FAULT 08:	The type of device is changed. **This is indicated only once and will disappear after resetting the warning.**
FAULT 09:	New addresses (devices) are found in the loop. **This is indicated only once and will disappear after resetting the warning.**

8. Conventional detector loops

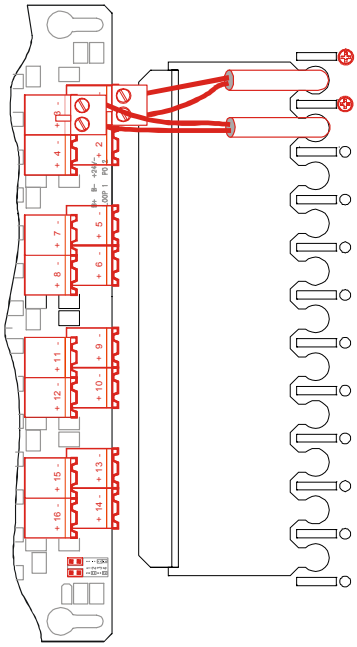
8.1 Measuring the Cables

1. Disconnect all power from the control panel.
2. Measure the resistance between the wires of the conventional line cable. Set the meter to low voltage resistance measurement (not diode measurement)
   Depending on connected devices the result should be:
   - If all devices are normally open (NO) types and no EOL resistor is connected the meter should indicate a break in the line.
   - If all devices are normally open (NO) types and an EOL resistor is connected the meter should indicate the value of the EOL resistor.
   - If some devices are normally closed (NC) types the meter should indicate the parallel connection of the series resistances of those devices.
   If the resistance meter indicates other values, the cause of the fault must be located. (It may be a detector connected the wrong way or a short circuit in the conductors).
3. Short-circuit the wires of the conventional detection line cable at the control panel during measuring.
4. Disconnect the end-of-line resistors from those conventional detection line terminals to which the cables are to be connected.
5. Take the end-of-line resistor with you and walk to the end of the line and measure the resistance between the wires of the cable. The maximum allowed resistance is 100Ω (50Ω if the line goes through an Exi barrier). If the resistance is higher, there is a break in the line (did you remember short-circuit the cable wires at the control panel). Locate the breaks and eliminate them.
6. Connect the end-of-line resistors to their respective places (the last detector or manual call point of the line or the end-of-line unit box connected after the last device).

8.2 Connecting the loop to the control panel

Each CLC board has 16 conventional lines (detection circuits).

1. Disconnect all power from the control panel.
2. If shielded cable is used, connect the shield to the nearest earth screw in the back plate.
3. Connect the pair of wires of the conventional line to the terminals of the CLC board retaining polarity.



8.3 Functional check of the loop

1. Start the control panel by switching the power on	First connect the transformer cable, then the battery cable
2. Wait for about 3 minutes, until the panel has started up and scanned all devices connected to the loops.	The text 'Panel starting up' disappears from the bottom of the display.

The detector loop is in order if the control panel does not give fire, fault or service indications. Otherwise the fault must be located by following the instructions below.

3. If the panel indicates fire alarm, press the "BUZZER SILENCE" button, scroll the alarm indications with the "MORE ALARMS"- button and note them.
4. Eliminate the causes of fire alarm and reset the indications by pressing the "RESET"- button until the pulsed signal stops.
5. Repeat the above mentioned procedures until there are no more fire alarms.
6. If the panel indicates fault or maintenance warning press the "BUZZER SILENCE" button, scroll the indications with the "MORE ALARMS"- button and note them.
7. Eliminate the causes of fault and maintenance warnings and reset the indications by pressing the "RESET"- button until the pulsed signal stops.
8. Repeat the above mentioned procedures until there are no more fault and maintenance warnings.
9. Test that each detector and manual call point gives an alarm. Test procedures for the various devices are given in the documentation for those devices.

Possible causes for fire alarm:

- A manual call point is pressed down; the glass has been broken or taken off.
- Smoke, water vapour or thick dust in a smoke detector.
- Heat close to a heat detector.

Possible causes for fault warnings:

- A break or a short circuit in the loop.
- An earth leak in the loop.
-

8.4 Compatible detectors and manual call points

Compatibility of detectors with the CLC conventional line is determined by the following factors:

- Supply voltage range
- Current consumption in standby condition
- Voltage across the detector in alarm condition
- Series resistance (either in the detector or in the base)
- End-Of-Line resistor

The voltage supplied by the CLC to the conventional line is 21Vdc to 24Vdc. The maximum allowable voltage drop in the cable is 21V minus the lowest operating voltage of the connected devices.

If the line goes through an Exi barrier, the maximum allowed cable resistance and current consumption is less than for a normal line.

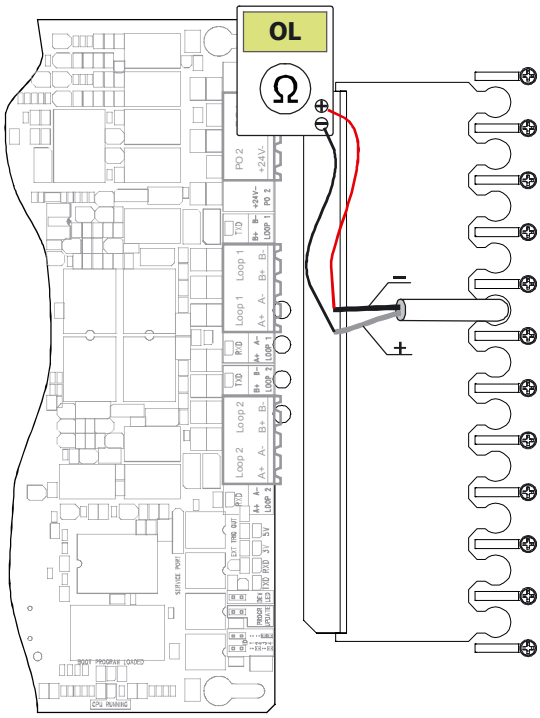
The following table shows the required series resistor for a number of detector voltages (in alarm condition), the two allowed EOL types and whether or not a Exi barrier is connected to the loop.

EOL resistor, Exi	4k7, 5%, not Exi	2k94, 1%, not Exi	4k7, 5%, Exi	2k94, 1%, Exi
Max cable resistance	100Ω	100Ω	50Ω	50Ω
Max detector load	1,8 mA	4,0 mA	1,5 mA	3,0 mA
8V	50 – 1000Ω	50 - 550Ω	10 - 700Ω	10 - 320Ω
5V	110 - 1300Ω	110 - 750Ω	150 - 1050Ω	170 - 550Ω
3V	140 - 1500Ω	150 - 880Ω	250 - 1250Ω	280 - 710Ω
1V	180 - 1700Ω	190 - 1010Ω	340 - 1500Ω	380 - 880Ω
0V	200 - 1800Ω	210 - 1070Ω	390 - 1600Ω	440 - 960Ω

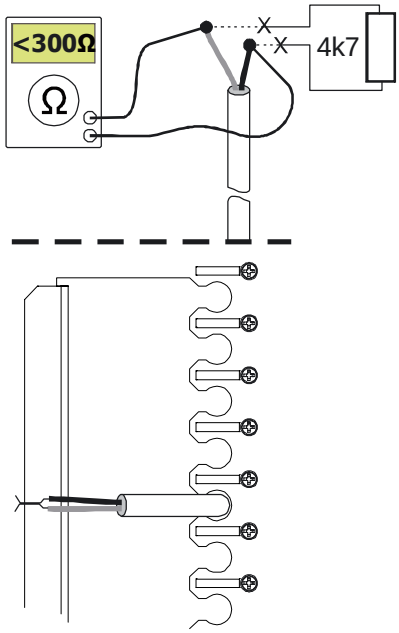
9. Alarm device lines

9.1 Measuring the cables and connecting the end-of-line resistors

1. Disconnect all power from the control panel.
2. Measure the resistance between the alarm device line cable wires so that the resistance meter positive (feeding) measuring end is in the "–" wire of the line. The resistance meter must indicate a break. If the resistance meter indicates a resistance value, the cause of the fault must be located. (It may be an alarm device connected the wrong way, a serial diode missing from the alarm device or a short circuit in the conductors).



3. Short-circuit the alarm device line cable wires at the control panel during measuring.
4. Disconnect the end-of-line resistors from those alarm device line outputs to which the cables are to be connected.
5. Measure the resistance between the wires of the cable at the last alarm device or end-of-line unit box in each alarm device line with the resistance meter. The allowed resistance should not be more than 300Ω depending on the current consumption of the alarm devices connected to the same line (*Fig 10*). If the resistance is considerably higher, there is a break in the line (did you remember short-circuit the cable wires at the control panel). Locate the breaks and eliminate them.
6. Take the end-of-line resistors to their respective places (the last alarm device of the line or the end-of-line unit box connected after the last alarm device) and connect them.



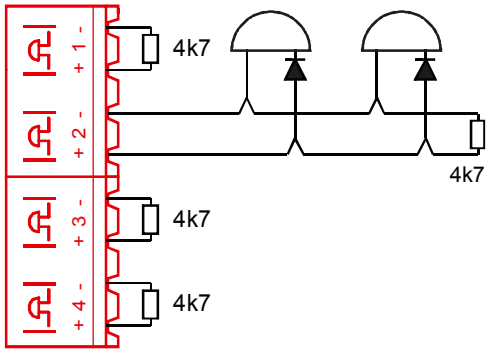
Note!
There must not be other resistors or end-of-line units in the alarm device line

9.2 Connecting the alarm device lines to the control panel

The FX-MC board has one alarm device line connector
The FX-IOC board has four alarm device line connectors

1. Check that all power has been disconnected from the control panel.
2. Connect the alarm device line cable wires to the terminals as follows:

- The fire alarm line is connected to the output marked with bell symbol on the MC and the outputs 1 to 3 marked with the bell symbol on the IOC.
- The fault alarm line is connected to the output 4 marked with the bell symbol on the IOC.
- 4,7kΩ resistors are left in the terminals of outputs that are not in use.



Note!

The above usages of the outputs are valid for non-configured panels. If configured, the purpose may have been changed. Verify with the configuration.

9.3 Functional check of alarm device lines

1. Start the control panel. The alarm device lines are in order if the control panel does not give fault indications.
2. If the alarm devices give fault indications, press the “BUZZER SILENCE”- button on the user panel.
3. You may also want to silence fault warning devices by pressing the “SILENCE/RESOUND” button.

Possible causes of fault indications:

- A short circuit in the line, the end-of-line resistor is too small (should be 4,7kΩ), a serial diode is missing from an alarm device or the device has been connected with the wrong polarity.
- A break in the line, the end-of-line resistor is missing or its resistance is too big.
- An earth leak in the line.

4. Eliminate the causes for fault indication and reset the panel by pressing the “RESET”- button until the audible signal stops.
5. Test the alarm device line operation as follows:

Note!

Ensure that all persons in the building have been informed of the alarm device test.

6. Press the wheel to activate the menu and select: CONTROLS
7. Then press the wheel again and select: FIRE ALARM DEVICES CONTROL
8. Press the “TEST”- button, at which the display will show the text: PRESS WHEEL TO ACTIVATE THE CONTROL
   - Pressing the wheel activates all fire alarm devices. ACTIVE
9. The test is stopped by pressing the “TEST”- button. Check the operation of the alarm devices. Eliminate any faults and repeat the test until all alarm devices operate

10. Control outputs

10.1 Clean relay outputs

The FX - MC board has two clean contact outputs
Each FX - IOC board has two clean contact outputs

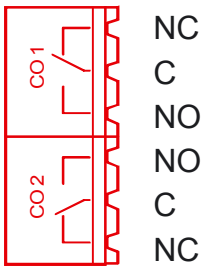
The function of the outputs is defined with the configuration tool.
The default functions of the outputs of a control panel that has not been configured are:

MC-CO1	Fire alarm router activation	
MC-CO2	Fault warning router activation	The relay is energized in normal condition and releases in fault warning condition or if power is removed
IOC-CO1	Fire alarm control output	
IOC-CO2	Fire alarm control output	

1. Disconnect all power from the control panel.
2. Select suitable outputs and connect the device to be controlled to the output.
   The relay contacts are rated for max. 30 VDC, 1A.

WARNING!

Voltage 230VAC must not be brought to the relay contacts.
If a device with 230 VAC is to be controlled, a suitable intermediate relay must be used. It is to be placed in a casing outside the control panel cabinet and equipped with a protective diode (e.g. 1N4005).



Test the fire outputs control operation as follows:

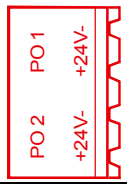
1. Start the control panel.
2. Press the wheel to activate the menu and select: CONTROLS
3. Then press again the wheel and select: FIRE OUTPUTS CONTROL
4. Press the "TEST"- button, at which the display will show the text: PRESS WHEEL TO ACTIVATE THE CONTROL
   - Pressing the wheel activates all control outputs. ACTIVE
5. The test is stopped by pressing the "TEST"- button.

10.2 Free power outputs

The power supply is normally 27 VDC, max.500 mA. During a power failure the supply follows the battery voltage. The power supply may momentarily be 30 VDC at the most.

IOC, LC and ALCB, PO1 and PO2
ALCA only PO1
MC, only PO

PSA , PO1 and PO2; 2A
PSB , PO1 and PO2; 4A



NOTE! The max. current consumption of the **FX** - control panel, all loops and addresses and all control panel outputs is 1 A in standby condition and 4.5 A in alarm condition. The standby time required for the system may limit the	NOTE! The max. current consumption of the **FXM** - control panel (with PSA- power supply board), all loops and addresses and all control panel outputs is 0.5 A in standby condition and 2.2 A in alarm condition. The

max. load of the outputs.	standby time required for the system may limit the max. load of the outputs.

1. Start the control panel.
2. If the control panel gives a fault indication press the “BUZZER SILENCE”- button. You may also want to press the SILENCE/RESOUND button to silence fault warning devices.
3. If the control panel gives an earth fault indication, the cause may be an earth leak in the cable or an earth leak caused by the device to be supplied with power.
4. Power output fault indication means overload or short circuit in the output.
5. Disconnect all power from the control panel, repair any fault and restart the control panel.

11. Signal inputs

11.1 Connecting the inputs to the control panel

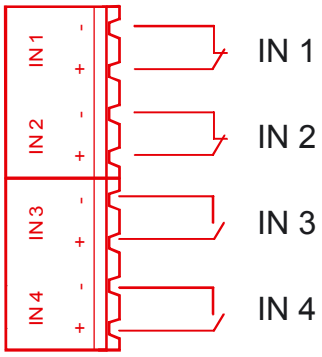
The FX-MC board has two clean contact input connectors
Each FX-IOC board has four clean contact input connectors

A clean contact, for example relay points or a switch, can be connected to the inputs.

The function and normal state (normally open or normally closed) of the inputs is defined with the configuration tool.

The functions of the inputs of a control panel that has not been configured are:

Input	Function	Normal state
MC-IN1	Fault in fire alarm router equipment	Normally closed
MC-IN2	Fault in fault warning router equipment	Normally closed
IOC-IN1	Fault warning	Normally closed
IOC-IN2	Fault in extinguisher	Normally closed
IOC-IN3	Extinguisher activated	Normally open
IOC-IN4	Smoke vents activated	Normally open



1. Disconnect all power from the control panel.
2. Select suitable inputs and connect the device to be monitored (e.g. extinguisher call point, router fault output) to the control panel.
3. Start the control panel.
4. If the control panel gives an earth leak indication, press the “BUZZER SILENCE”- button and disconnect all power. Repair any fault and restart the control panel.
5. Test the operation of the inputs with the connected device (remember that a fire alarm or fault warning also activates fire and fault alarm devices and alarm routing equipment).

11.2 Testing the inputs

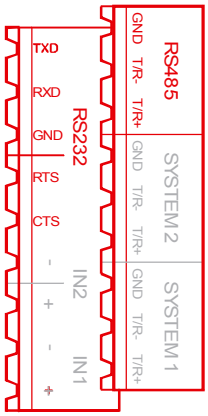
The inputs for Fault in fire alarm router, Fault in fault warning router and Fault in extinguisher can be tested without activation of fault warning outputs, as follows:

1. Press the wheel to activate the menu and select:	MONITORINGS
2. Then press again the wheel and select:	FIRE ROUTING MONITOR
3. Press the “TEST”- button, at which the display will show the text:	IN TEST STATE
• The display indicates now the status of the input	FAULT or OK

4. The test is stopped by pressing the “TEST”- button.

12. Serial communication ports

- The RS232 communication port is standard on the FX-MC board
- The RS485 communication ports require that the FX-SA board is installed on the FX-MC board
- The function of the serial communication ports is defined with the configuration tool.
- If the control panel is not configured, the communication ports are not in use.



SAA 485 (INFO)

SAB 485 (INFO), SYSTEM1, SYSTEM2

1. Disconnect all power from the control panel.
2. Connect the RS485 serial communication wires to the terminals RS485 as follows:

   Data wires to terminals T/R + and -. GND wire to terminal GND.

   The shield has to connect to the control panel cabinet through the parallel connected capacitor and varistor as follows (because long distances may cause big differences in earth potential). Loosen three screws of the backplate. Install the small circuit board from the SA package under the screws and tighten the screws and connect the shield to the terminals.
3. Connect the RS232 serial communication conductors to the RS232 terminals as follows:

   The control panel transmitter wire to terminal TxD

   The control panel receiver wire to terminal RxD

   GND wire to terminal GND.

   Connect the shield to the control panel backplate.
4. Start the control panel.
5. If the control panel gives an earth leak indication, press the “BUZZER SILENCE”- button and disconnect all power from the control panel. Repair any fault and restart the control panel.
6. If the control panel has been configured, the operation of serial communication should be tested (remember that a fire or fault also activates fire or fault alarm devices and alarm routing equipment).

13. Alarm routing equipment

Connecting the alarm routing equipment to the control panel
The alarm routing equipment must not be connected until all other necessary commissioning procedures have been performed.

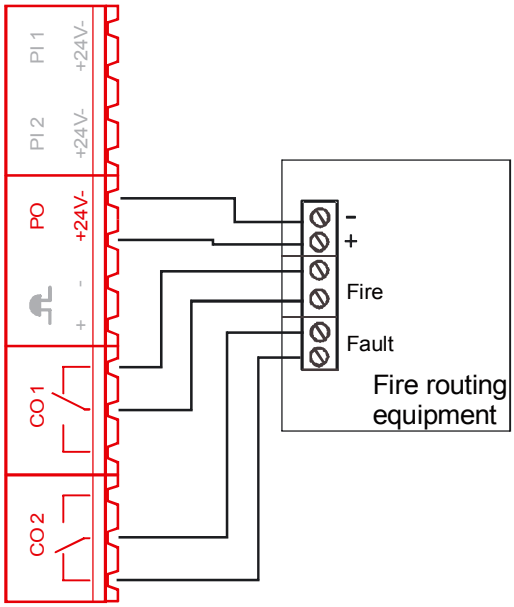
> **Note!**
> **Inform the alarm receiver about the alarm routing equipment test.**

1. **Disconnect all power from the control panel.**
2. **Connect the alarm routing equipment to the terminals CO1, CO2 and PO on the MC board.**

The default function of CO1 and CO2 are:

- **CO1 = Fire Alarm Router**
- **CO2 = Fault Warning Router**
- **The PO terminal provides 24 Vdc supply for the routing equipment.**

Note!
CO2 is energized (as drawn) when the panel is in normal condition



3. **Start the control panel.**

- **If the control panel gives a fault indication, press the "BUZZER SILENCE"- button.**
- **If the control panel gives an earth leak indication, the cause may be an earth leak in the cable or an earth leak caused by the alarm routing equipment.**
- **If the control panel gives a power output fault indication, the alarm routing equipment connected draws too much current and is probably faulty or the connection is wrong.**
- **Disconnect all power from the control panel, repair any fault and restart the control panel.**

Test the alarm transmission (combined test of fault warning router and fire alarm router) operation as follows:

4. **Press the wheel to activate the menu and select:**	**ALARM TRANSMISSION TEST**
5. **Press the "TEST"- button, at which the display will show the text:**	**PRESS WHEEL TO ACTIVATE THE TEST**
6. **Pressing the wheel activates the fault warning router.**	**ALARM TRANSMISSION TEST STAGE 1** **Fault routing activated**
7. **After 30 seconds also the fire alarm router is activated.**	**ALARM TRANSMISSION TEST STAGE 2** **Fire routing activated**
8. **After a further 30 seconds the test stops automatically. The test can also be stopped at any time by pressing the "TEST" – button.**	**ALARM TRANSMISSION TEST**

14. Configuration

Set control panel to access level 3

1. Select with “MENU”-button from main menu
2. Press the wheel
3. Select the first digit with the wheel and press the wheel to enter
4. Repeat with the other digits
5. Press “ENABLE” when the right code is entered
6. The panel is now in access level 3

CURRENT ACCESS LEVEL:2
CHANGE TO ACCESS LEVEL 3:0000

CURRENT ACCESS LEVEL: 3

Set panel to configuration state

7. Set control panel to the configuration state by putting a jumper in position “CONF” on the MC board. The display shows the text:

CONFIGUARTION STARTED
READY FOR CONFIGURATION

Transfer the configuration

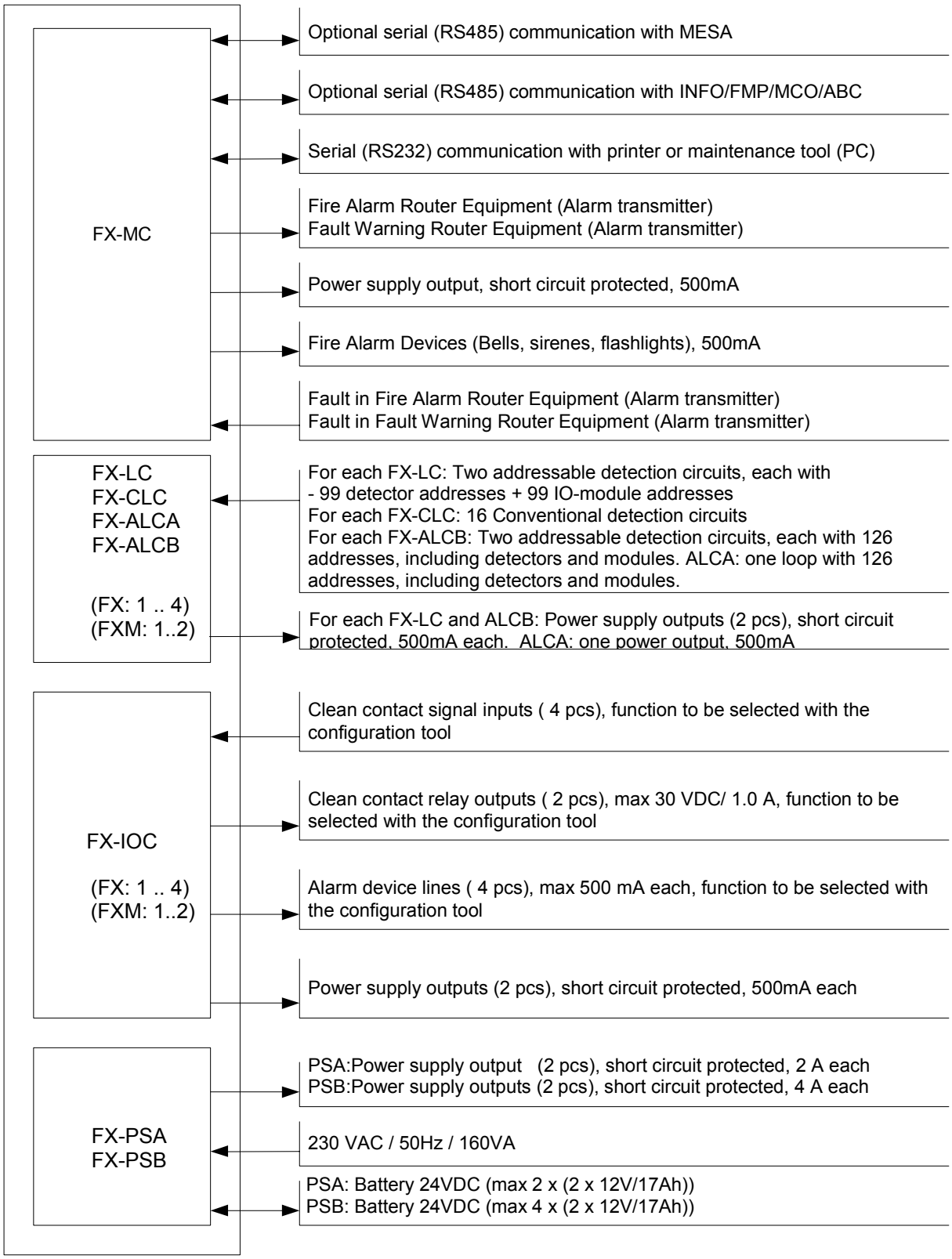
8. Connect configuration cable to the terminal RS232.
9. Configure the control panel with the configuration tool
10. Remove the configuration cable from the terminal RS232 and the jumper from the position “CONF” on the MC

Restart the panel

11. Restart the control panel by pressing “RESET” button on the MC.

15. Control panel connections

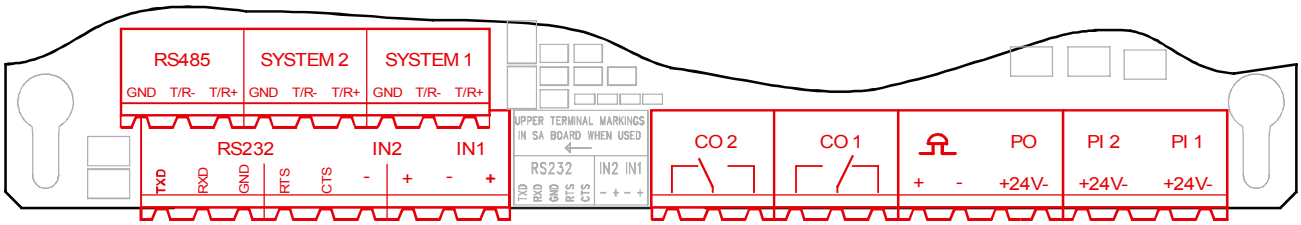
External connections



NOTE! The max. current consumption of the **FX** - control panel, all loops and addresses and all control panel outputs is 1 A in standby condition and 4.5 A in alarm condition. The standby time required for the system may limit the max. load of the outputs.	NOTE! The max. current consumption of the **FXM** - control panel (with PSA- power supply board), all loops and addresses and all control panel outputs is 0.5 A in standby condition and 2.2 A in alarm condition. The standby time required for the system may limit the max. load of the outputs.

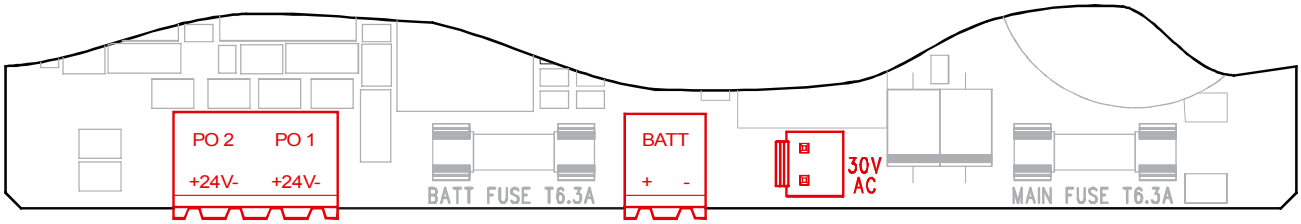
15.1 Connectors on the MC board

Terminal name	Connector symbol	Purpose	Description
PI 1	24 V -	Power supply 1 in (-)	For external power supply only. Not to be used if the panel is equipped with an internal power supply unit.
	24 V +	Power supply 1 in (+)	
PI 2	24 V -	Power supply 2 in (-)	For external power supply only. Not to be used if the panel is equipped with an internal power supply unit.
	24 V +	Power supply 2 in (+)	
PO	24 V -	Power supply out (-)	Power supply out for auxiliary equipment
	24 V +	Power supply out (+)	
	-	Alarm device line (-)	General fire alarm device line as default. Other functions can be selected with the configuration tool.
	+	Alarm device line (+)	
CO 1		Relay output 1 NC	Fire alarm router output as default.
		Relay output 1 C	
		Relay output 1 NO	
CO 2		Relay output 2 NO	Fault warning router output as default. The relay is energized in normal condition (as drawn) and releases in fault condition
		Relay output 2 C	
		Relay output 2 NC	
IN 1	+	Clean contact input 1	Fault in fire alarm router as default function.
	-		
IN 2	+	Clean contact input 1	Fault in fault warning router as default function.
	-		
RS232	CTS	Clear To Send	Serial communication with printer or configuration tool
	RTS	Request To Send	
	GND	Ground	
	RxD	Received data	
	TxD	Transmit data	
Following connectors are in use only if the SA board is mounted			
SYSTEM 1	T/R +	Transmit/Received Data +	Serial communication with MESA panels
	T/R -	Transmit/Received Data -	
	Gnd	Ground	
SYSTEM 2	T/R +	Transmit/Received Data +	Currently not in use
	T/R -	Transmit/Received Data -	
	Gnd	Ground	
RS485	T/R +	Transmit/Received Data +	Serial communication with INFO/FMP/MCO/ABC or other alarm management systems
	T/R -	Transmit/Received Data -	
	Gnd	Ground	



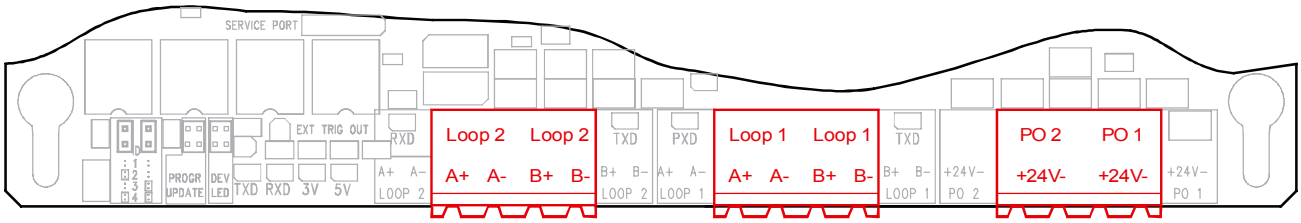
15.2 Connectors on the PSA and PSB- board

Terminal name	Connector symbol	Purpose	Description
30 VAC			30 VAC input from transformer
BATT	-	Battery -	Connection for the standby battery
	+	Battery +	
PO 1	24 V -	Power supply 1 out -	Power supply output 1 for external load
	24 V +	Power supply 1 out +	
PO 2	24 V -	Power supply 2 out -	Power supply output 2 for external load
	24 V +	Power supply 2 out +	



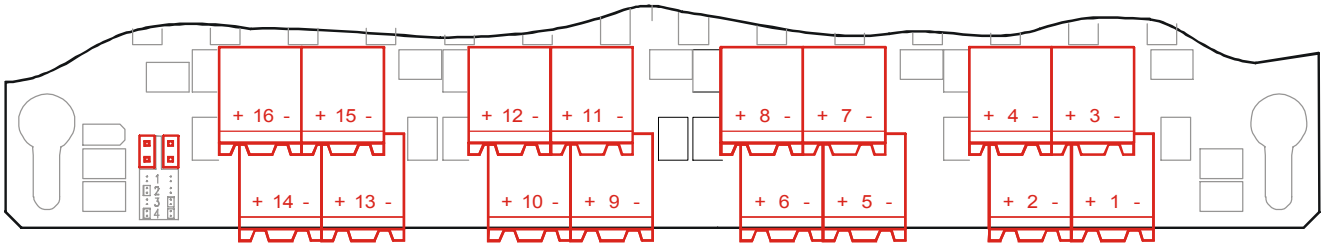
15.3 Connectors on each LC board

Terminal name	Connector symbol	Purpose	Description
PO 1	24 V -	Power supply out (-)	Power output for loop devices in loop 1 that require external power supply
	24 V +	Power supply out (+)	
PO 2	24 V -	Power supply out (-)	Power output for loop devices in loop 2 that require external power supply
	24 V +	Power supply out (+)	
LOOP 1	B -	Return end -	System Sensor- loop devices
	B +	Return end +	
	A -	Outgoing end -	
	A +	Outgoing end +	
LOOP 2	B -	Return end -	System Sensor- loop devices
	B +	Return end +	
	A -	Outgoing end -	
	A +	Outgoing end +	



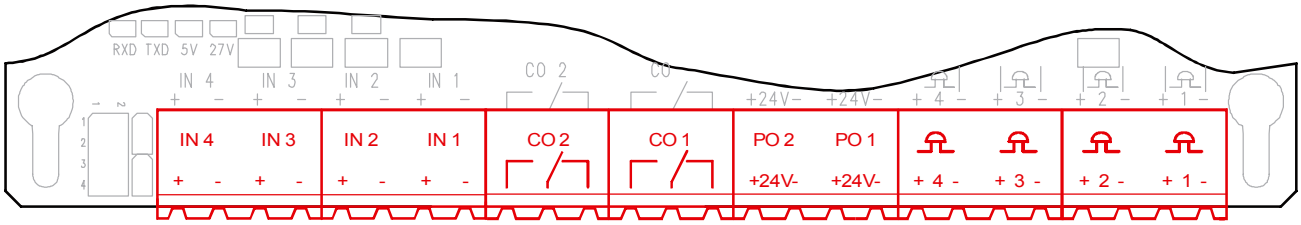
15.4 Connectors on each CLC board

Terminal name	Connector symbol	Purpose	Description
1	-	Conventional line 1 -	
	+	Conventional line 1 +	
...			
16	-	Conventional line 16 -	
	+	Conventional line 16 +	



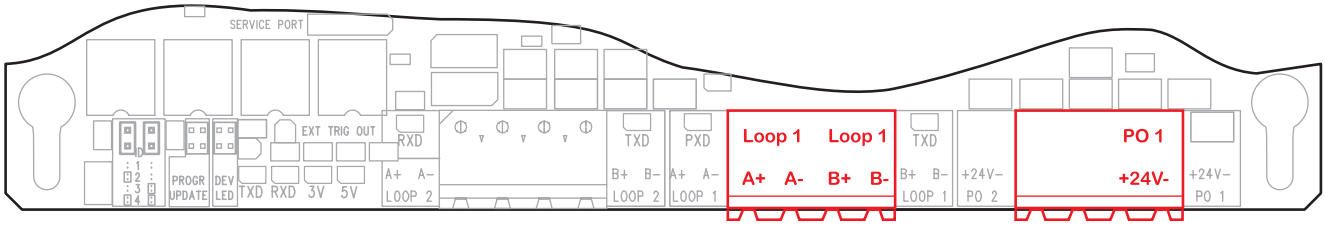
15.5 Connectors on each IOC board

Terminal name	Connector symbol	Purpose	Description
1	- +	Alarm device line 1 -	Fire alarm device line as default
		Alarm device line 1 +	
2	- +	Alarm device line 2 -	Fire alarm device line as default
		Alarm device line 2 +	
3	- +	Alarm device line 3 -	Fire alarm device line as default
		Alarm device line 3 +	
4	- +	Alarm device line 4 -	Fault warning device line as default
		Alarm device line 4 +	
PO 1	24 V -	Power supply 1 out -	Power supply output 1
	24 V +	Power supply 1 out +	
PO 2	24 V -	Power supply 2 out -	Power supply output 1
	24 V +	Power supply 2 out +	
CO 1		Relay output 1 NC	Clean contact relay output. General fire alarm output as default.
		Relay output 1 C	
		Relay output 1 NO	
CO 2		Relay output 2 NC	Clean contact relay output.
		Relay output 2 C	
		Relay output 2 NO	
IN 1	-	Clean contact input 1	
	+		
IN 2	-	Clean contact input 2	
	+		
IN 3	-	Clean contact input 3	
	+		
IN 4	-	Clean contact input 4	
	+		



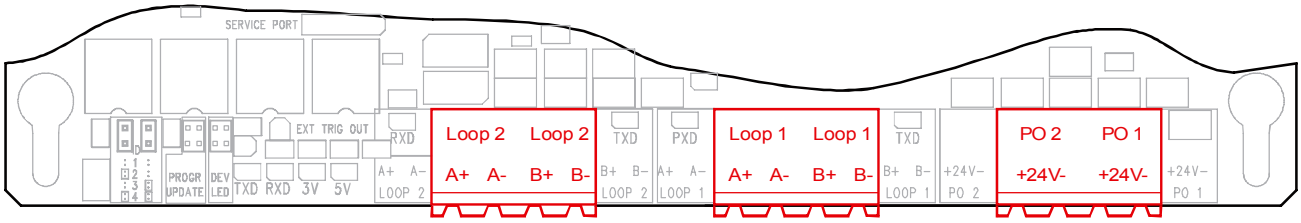
15.6 Connectors on each ALCA- board

Terminal name	Connector symbol	Purpose	Description
PO 1	24 V -	Power supply out (-)	Power output for loop devices in loop 1 that require external power supply
	24 V +	Power supply out (+)	
LOOP 1	B -	Return end -	Intellia loop devices
	B +	Return end +	
	A -	Outgoing end -	
	A +	Outgoing end +	



15.7 Connectors on each ALCB- board

Terminal name	Connector symbol	Purpose	Description
PO 1	24 V -	Power supply out (-)	Power output for loop devices in loop 1 that require external power supply
	24 V +	Power supply out (+)	
PO 2	24 V -	Power supply out (-)	Power output for loop devices in loop 2 that require external power supply
	24 V +	Power supply out (+)	
LOOP 1	B -	Return end -	Intellia loop devices
	B +	Return end +	
	A -	Outgoing end -	
	A +	Outgoing end +	
LOOP 2	B -	Return end -	Intellia loop devices
	B +	Return end +	
	A -	Outgoing end -	
	A +	Outgoing end +	



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